Chiyoda: Entropy-Guided Information Control for Hazard-Coupled Pedestrian Evacuation

A Simulation Framework for Measuring When Emergency Communication Helps or Harms

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Abstract

Emergency evacuation is usually modeled as a physical routing problem, with information appearing as a fixed assumption. We present *Chiyoda*, an Information-Theoretic Evacuation Dynamics framework that treats emergency communication as a controllable intervention in a coupled physical-information system. Agents carry probabilistic beliefs, exchange distorted information through local gossip, receive beacon and responder messages, suffer physiological impairment under hazards, and choose routes through belief-weighted path planning. The framework compares static, global, responder-relay, entropy-targeted, density-aware, exposure-aware, and bottleneck-avoidance policies. The central claim is not that more information is always better, but that emergency communication has measurable safety trade-offs.

1 Introduction

Emergency communication is usually treated as a benign input to evacuation: give people better information and their decisions should improve. That assumption is plausible at the individual level, but it is not obviously true at the crowd level. A warning that reduces uncertainty for everyone at once can also synchronize route choice, push agents toward the same exit, amplify a bottleneck, or move people through a contaminated region before the physical hazard has dissipated. The question is therefore not simply whether information helps, but when, where, and for whom it helps.

This paper studies evacuation as a coupled physical-information system. The physical side includes layout geometry, social-force movement, bottleneck dynamics, hazard spread, and physiological impairment. The information side includes partial exit knowledge, hazard beliefs, signage and PA broadcasts, responder credibility, local gossip, message decay, and distortion. Chiyoda connects these layers by making agents plan through their believed world rather than the ground truth world. They may avoid a hazard they have heard about, miss an exit they do not know, herd when uncertainty is high, or follow a stale route because the belief has not yet decayed.

Research question. We ask whether entropy-guided communication can improve evacuation outcomes without creating harmful convergence. More concretely: which intervention policies reduce belief entropy and improve belief accuracy per unit of induced exposure, queueing, and exit imbalance? The current study scaffold compares static signage, global broadcasts, responder relay, entropy-targeted broadcasts, density-aware targeting, exposure-aware targeting, and bottleneck-avoidance targeting.

Contributions. This paper makes four contributions.

1. **A controllable information-intervention layer.** Chiyoda turns emergency communication into a first-class policy object rather than a fixed scenario assumption. Policies can be static, authority-driven, or adaptive to entropy, density, exposure, and bottleneck pressure.

2. **A coupled ITED simulation loop.** Agents carry probabilistic beliefs over exits and hazards, update them through observation, beacon broadcast, responder relay, and gossip, then route through belief-weighted costs under social-force dynamics and hazard impairment.

3. **Information-safety metrics.** We introduce study outputs that jointly measure entropy reduction, belief accuracy gain, intervention reach, hazard exposure pressure, queue pressure, exit imbalance, and a harmful convergence index.

4. **A reproducible paper pipeline.** YAML study definitions generate multi-seed bundles with Parquet/CSV tables and paper figures. The LaTeX paper ingests generated study statistics through `stats.tex`.

Roadmap. §2 positions the work against pedestrian dynamics, information-aware evacuation, and ABM validation work. §3 describes Chiyoda's model and intervention policies. §4 explains the software pipeline. §5 lays out the planned empirical study. §6 sharpens the distinction from prior work. §7 states the current threats to validity, and §8 concludes.

2 Background

Pedestrian evacuation models have a long physical tradition. The social-force model represents pedestrians as self-driven particles subject to destination, repulsion, and contact forces [2]. Cellular automata models discretize space and update local movement decisions through transition rules [1]. Fundamental-diagram work relates crowd density to walking speed and flow capacity [9]. These tools are useful because evacuation failure is often physical: bottlenecks, counter-flow, density waves, and slower-is-faster effects can dominate individual intent.

Hazard-coupled evacuation adds another layer. In CBRN or smoke scenarios, the environment changes while people move through it; visibility, speed, and physiology can degrade as a function of exposure. Prior toxic-gas evacuation models already show that information perception and transmission matter for route choice under a spreading contaminant [4, 3]. Chiyoda builds on this insight but makes the communication policy itself the experimental object.

Information-aware evacuation models also intersect with game theory and bounded rationality. Bayesian formulations can model incomplete information and route-choice equilibria under congestion [8]. Social group ABMs review the importance of communication, affiliation, and collective behavior, while also noting persistent gaps in behavioral realism and validation [7, 5]. Chiyoda takes a simulation-first position: it does not claim to solve empirical validation in one paper, but it makes belief uncertainty measurable and interventions reproducible.

The information-theoretic lens comes from Shannon entropy [6]. For each agent, Chiyoda computes entropy over exit existence, hazard knowledge, and general danger. Global entropy is the population mean. Entropy is not a normative objective by itself: the point of this paper is to test when reducing entropy improves safety and when it merely concentrates the crowd's behavior.

3 Design

Chiyoda models evacuation as a feedback loop between physical state, information state, and action. At each step, hazards evolve; agents observe nearby exits and hazards; beacons, responders, gossip, and intervention policies update beliefs; agents revise intentions; belief-weighted navigation selects paths; social-force dynamics move agents; and telemetry records the macroscopic outcome.

3.1 Belief State

Each agent carries a belief vector over exits, hazards, and general danger. Exit beliefs store existence probability, congestion estimate, freshness, source credibility, and gossip hop count. Hazard beliefs store position, severity estimate, radius estimate, freshness, credibility, and hop count. Agents do not route through the ground truth environment directly; they route through the subset of exits and hazards represented in their beliefs.

3.2 Communication Channels

The baseline ITED channels are direct observation, beacon broadcast, and agent-to-agent gossip. Observation is range-limited by visibility and physiological impairment. Beacons provide high-credibility exit knowledge within a fixed radius. Gossip is local, probabilistic, and distorted by hop count and panic state.

3.3 Intervention Policies

The new contribution is an intervention policy interface. Every policy decides whether to fire, selects one or more spatial targets, builds a message, applies it to recipients inside a radius, and emits telemetry. Chiyoda currently implements seven policies:

- **Static beacon**: periodic local messages from existing signage or PA beacon locations.
- **Global broadcast**: station-wide high-reach communication.
- **Responder relay**: mobile high-credibility messages emitted by released responders.
- **Entropy-targeted**: targets agents with highest belief entropy.
- **Density-aware**: targets dense local clusters.
- **Exposure-aware**: targets agents under high current or cumulative hazard pressure.
- **Bottleneck-avoidance**: targets active bottleneck zones.

Figure 1 summarizes the policy space. Figure 2 shows where the intervention hook sits in the simulation loop.

3.4 Information-Safety Metrics

The core metric is information-safety efficiency:

$$\eta_{safe} = \frac{\Delta H^+ + \Delta A^+}{1 + P_{exposure} + P_{queue}},$$

where ΔH^+ is positive entropy reduction, ΔA^+ is positive belief-accuracy gain, $P_{exposure}$ is recipient-weighted hazard-load pressure, and P_{queue} is bottleneck pressure observed at intervention time. Chiyoda also records intervention reach, exit imbalance, and a harmful convergence index that rises when entropy reduction coincides with concentrated exit usage, queueing, and exposure.

4 Implementation

Chiyoda is implemented as a Python research toolkit. Scenario YAML files define layouts, cohorts, hazards, responders, information parameters, intervention policies, seeds, and variants. The scenario manager builds

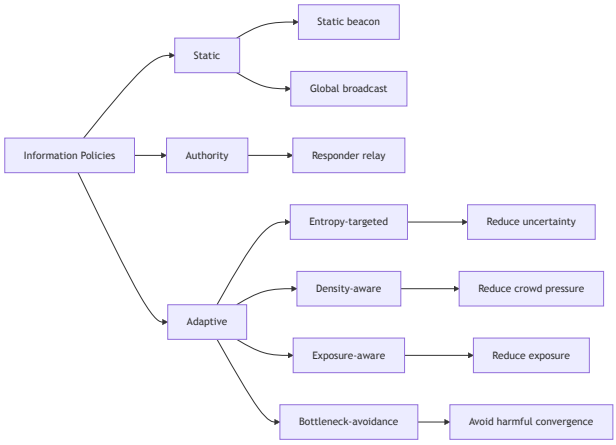

Figure 1: Information intervention policy taxonomy. Policies differ by source, target signal, spatial reach, and safety objective.

Figure 2: Intervention hook in the ITED simulation loop. Policies read the current physical-information state, update recipient beliefs, and emit before/after telemetry before navigation.

a simulation by loading the layout, constructing agents, hazards, exits, responders, navigation, spatial indexing, behavior state machines, and optional intervention policies.

The core implementation lives in `chiyoda/core/simulation.py`. The simulation loop records step telemetry, agent telemetry, cell-level maps, exit flow, bottleneck dynamics, hazard state, gossip events, and intervention events. Study execution lives in `chiyoda/studies/runner.py`; it materializes variants, runs each seed, aggregates tables, and exports study bundles. Analysis and figure export live in `chiyoda/analysis/`.

The information intervention layer lives in `chiyoda/information/interventions.py`. It is deliberately small: policies produce target points and messages, while the shared executor applies messages to recipients and computes before/after entropy and accuracy. This keeps policy design separate from belief mutation and telemetry.

Two implementation details matter for reproducibility. First, A* pathfinding uses per-step caches for density penalties, edge weights, and identical start/goal/belief queries. These caches are cleared whenever the simulation advances to a new navigation phase, so they do not change the cost function; they only avoid recomputing identical values during the same step. Second, gossip reports about the same physical hazard are merged into an existing hazard belief rather than appended as duplicate rumors. This is both a performance requirement and a modeling requirement: repeated reports should update credibility, freshness, severity, and radius estimates for the same hazard, not create an unbounded list of separate hazards.

The paper pipeline mirrors the code pipeline. Running `python3 -m chiyoda.cli sweep scenarios/study_information_control.yaml` emits tables and figures; `paper/scripts/gen_stats.py` converts the exported bundle into `stats.tex`; and the LaTeX build injects those values into the abstract and evaluation.

[TODO] *Replace this implementation sketch with line-referenced architecture details once the model stabilizes.*

5 Evaluation

We evaluate Chiyoda on `study_information_control.yaml`, a five-seed comparison of eight emergency communication regimes: no deliberate intervention, static beacon broadcast, global PA broadcast, responder relay, entropy-targeted communication, density-aware communication, exposure-aware communication, and bottleneck-avoidance communication. The exported bundle contains 40 runs, 8 variant aggregates, and 1803 intervention events. Each run uses the same station-sarin scenario, a 120-agent commuter/tourist population, asymmetric initial information, gossip,

beacon infrastructure, three delayed responders, and a 500-step horizon.

5.1 Outcome Measures

We report three classes of outcome. Physical outcomes include evacuated agents, mean travel time, incapacitation count, mean hazard exposure, peak bottleneck queue, and exit imbalance. Information outcomes include mean entropy, entropy reduction, intervention reach, entropy reduction attributable to interventions, and belief-accuracy gain. Coupled outcomes include information-safety efficiency and harmful convergence. These coupled metrics are the main object of the paper: they ask whether a policy buys useful belief improvement without creating exposure or queueing pressure.

5.2 Aggregate Findings

Table 1 summarizes the variant aggregates. The first result is negative but important: no intervention policy dominates the baseline across physical evacuation outcomes at this seed count. The no-intervention baseline evacuates 11.8 agents on average, while static beacons evacuate 12.4, responder relay evacuates 11.4, and the other adaptive policies evacuate between 9.4 and 10.6. Mann–Whitney tests against the baseline do not show significant improvement in evacuated count, travel time, incapacitation, or exposure for any policy at $n = 5$ seeds. This is exactly why the paper cannot simply claim "more information improves evacuation."

The second result is that communication policy matters even when physical outcomes remain noisy. Static beacons have the highest information-safety efficiency in this study, 0.018, and are the current best policy by that metric (static_beacon). They produce a modest evacuation increase over baseline (+0.6 agents), a small travel-time reduction (-0.29s), and lower peak queue (2.0 vs. 2.2), but the effect is not statistically decisive. Their strength is efficiency: local, repeated, high-credibility messages produce belief improvement without the full synchronization cost of global broadcast.

The third result is that wide reach is not the same as useful reach. Global broadcast reaches 1823.6 recipients on average across intervention events, far more than static beacons (492.6), but its information-safety efficiency is only 0.0057. It also has the lowest entropy reduction among intervention policies (0.044 aggregate entropy reduction versus 0.195 in the baseline trajectory). The model interpretation is that a broad message can homogenize beliefs without necessarily producing safer route allocations.

The fourth result is the clearest support for the paper's thesis: entropy-targeted communication improves belief accuracy strongly but can backfire at the system level. It produces the largest intervention accuracy gain (1.713), but also the highest peak bottleneck queue (3.6), the fewest evacuated agents (9.4), and the highest harmful-convergence index (8.62). The Mann–Whitney comparison against no intervention finds the queue increase borderline at $p = 0.048$ with a large effect size ($d = 1.57$). This is not a final causal proof, but it is a useful warning: reducing the uncertainty of the most uncertain agents can synchronize newly informed agents into the same capacity-constrained routes.

5.3 Hypotheses Revisited

H1 predicted that adaptive targeting would improve information-safety efficiency over global broadcast. The result is mixed. Entropy-targeted, density-aware, exposure-aware, and bottleneck-avoidance policies all have nonzero efficiency, but only bottleneck avoidance exceeds global broadcast by a substantial margin; entropy targeting and density targeting improve some information measures while increasing harmful convergence.

H2 predicted that global broadcast would reduce entropy quickly but risk route synchronization. The data support the second half more than the first. Global broadcast has high reach and high harmful convergence, but its aggregate entropy reduction is low. In this model, broadcast reach is partly wasted when messages arrive after agents have already converged or when congestion information is insufficiently differentiated.

H3 predicted that exposure-aware targeting would reduce hazard exposure. Exposure-aware policy has the lowest mean hazard exposure among intervention policies (3.140), slightly below baseline (3.147), but the difference is small and not significant. The result suggests exposure targeting is directionally reasonable but underpowered in the current scenario.

H4 predicted that responder relay would be robust but delayed. This is the strongest supported hypothesis: responder relay has the lowest mean hazard exposure among the high-efficiency policies (3.050), the lowest harmful convergence among intervention policies except exposure-aware targeting, and the second-best information-safety efficiency. Its limitation is timing: responders enter after the evacuation process has already begun.

5.4 Interpretation

The study supports the paper's central framing rather than a single winning policy. Information interventions should be evaluated as safety-control actions, not as monotonic entropy minimizers. A policy can reach many agents and still have low safety efficiency; another can improve belief accuracy and still worsen bottleneck pressure. The promising direction is therefore not "broadcast more," but "broadcast locally, credibly, and with awareness of capacity and exposure." Static beacon and responder relay policies currently look safest under this scenario; entropy targeting is scientifically interesting precisely because it exposes a failure mode.

5.5 Threats to Evaluation

These results are preliminary. Five seeds are enough to reveal qualitative tradeoffs and obvious failure modes, but not enough for final PhD-level claims about policy superiority. The next study should increase seed count, vary hazard severity and population familiarity, and

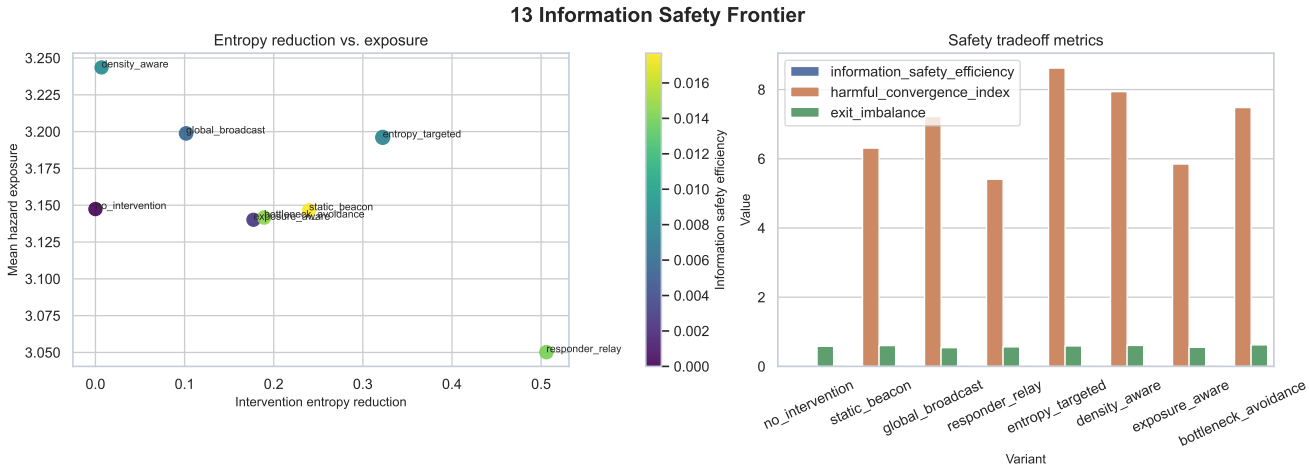

Figure 3: Information-safety frontier across communication policies. Static beacons produce the highest current information-safety efficiency, while entropy-targeted communication produces larger belief gains but the highest harmful-convergence index.

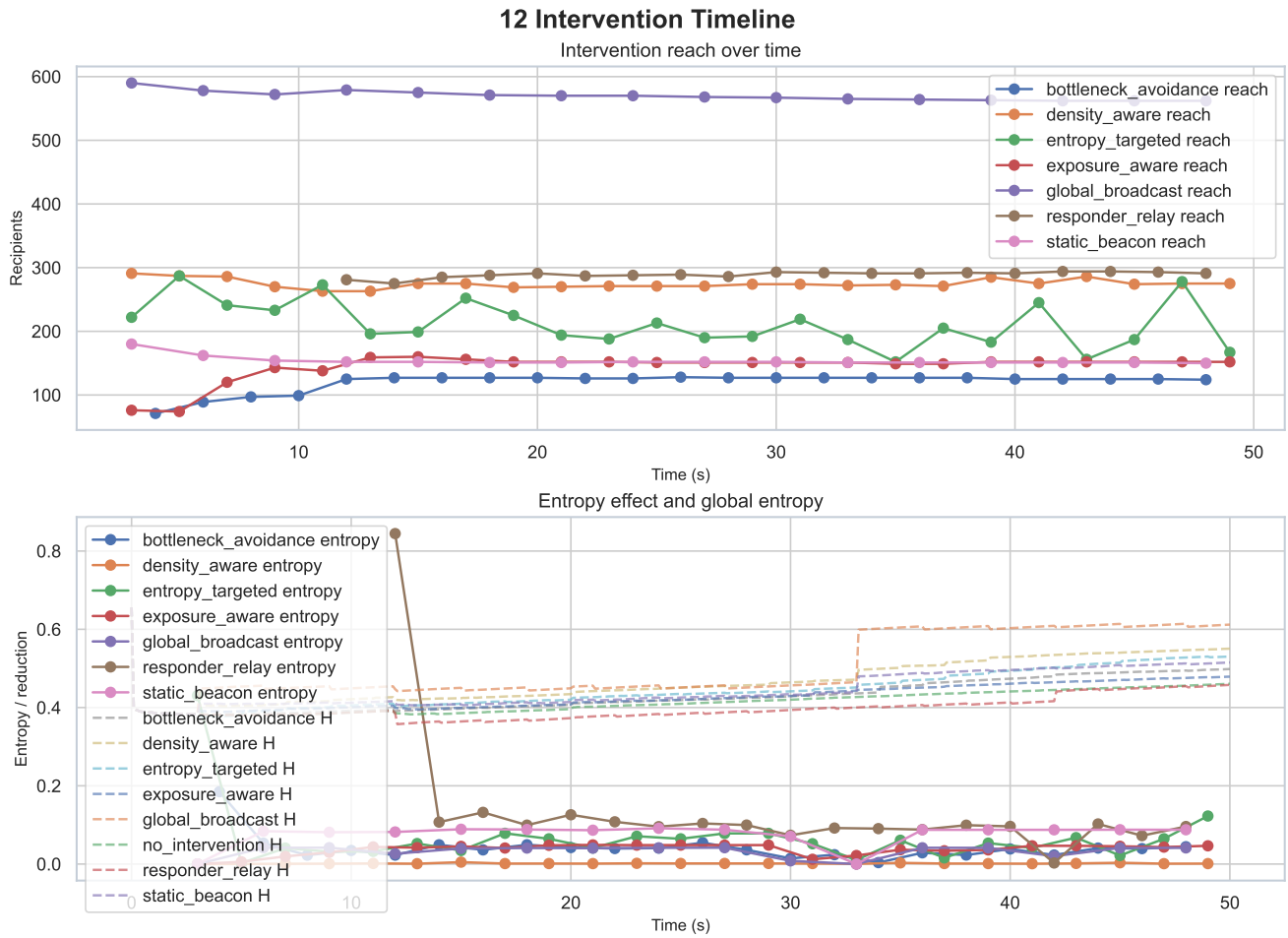

Figure 4: Intervention reach and entropy effects over time. Global broadcast reaches the most recipients, but wide reach is not equivalent to efficient safety improvement.

add calibrated geometries. The current metrics also depend on the chosen harmful-convergence formula; future work should test alternative normalizations and compare them against trajectory-level bottleneck events.

6 Related Work

Chiyoda sits between pedestrian dynamics, hazard-coupled evacuation, and information-aware decision models. Social-force and cellular automata models provide the physical base for movement and congestion [2, 1]. Toxic-gas evacuation work already shows that

Table 1: Variant aggregates from the full information-control study. Values are means across five seeds. ISE denotes information-safety efficiency; HCI denotes harmful-convergence index.

Policy	Evac.	Travel (s)	Exposure	Queue	ISE	HCI
No intervention	11.8	12.17	3.147	2.2	0.000	0.000
Static beacon	12.4	11.89	3.147	2.0	0.0177	6.31
Responder relay	11.4	11.81	3.050	2.4	0.0140	5.41
Bottleneck avoidance	10.0	11.48	3.142	2.4	0.0145	7.48
Density-aware	10.6	9.86	3.244	2.0	0.0082	7.94
Entropy-targeted	9.4	10.43	3.196	3.6	0.0078	8.62
Global broadcast	10.6	11.33	3.199	2.6	0.0057	7.22
Exposure-aware	10.6	10.83	3.140	2.0	0.0025	5.85

information perception and transmission affect evacuation outcomes under spreading hazards [4, 3]. Bayesian and game-theoretic models study incomplete information and route choice under congestion [8]. Social-group ABM surveys emphasize communication and collective behavior while documenting gaps in behavioral realism and validation [7, 5].

The distinction is that Chiyoda treats communication as an experimental control surface. It does not only ask whether agents possess information; it asks how an emergency system should allocate limited communication budget under uncertainty. This makes message timing, targeting, credibility, and spatial reach comparable interventions with measurable safety tradeoffs.

[TODO] *Expand this section after the final bibliography pass. In particular, separate empirical evacuation studies, ABM method papers, CBRN-specific evacuation papers, and information/control-theoretic routing papers.*

7 Limitations

Chiyoda is currently a stylized research simulator, not an operational evacuation planner. The station layouts are simplified grids unless imported from richer geometry. Hazard physics are intentionally lightweight and should not be read as validated CBRN dispersion models. Physiological impairment uses piecewise exposure curves that are useful for comparative simulation but not medical prediction.

The behavioral model is also a hypothesis generator rather than a calibrated description of a specific crowd. Agents use simplified BDI-style intentions, local gossip, panic/anxiety states, and familiarity cohorts. These mechanisms are valuable because they make assumptions inspectable, but the paper must not claim real-world predictive validity without calibration against evacuation drills, incident records, controlled VR experiments, or pedestrian trajectory data.

Finally, entropy is not equivalent to safety. The point of the paper is precisely that uncertainty reduction can be harmful when it produces synchronized movement. The proposed information-safety metrics are therefore comparative diagnostics, not universal welfare functions.

8 Conclusion

Chiyoda reframes evacuation as an information-control problem. In a hazard-coupled crowd, emergency communication changes beliefs, beliefs change routes, and route convergence changes exposure and congestion. The framework therefore measures not only whether information spreads, but whether spreading it at a particular time, place, credibility, and radius improves safety.

The repository now contains the code hooks, study definitions, metrics, and paper scaffold needed to evaluate that claim. The next step is empirical: run larger seeded studies, quantify uncertainty, identify regimes where global messages backfire, and refine adaptive policies that reduce entropy without creating harmful convergence.

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